

A self-contained volume bound for bodies of constant width

Circumsphere contacts, compulsory completion, and the constant 0.4144

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Abstract

We give a detailed computer-assisted proof that every convex body in three-dimensional Euclidean space of constant width $w > 0$ has volume strictly larger than $(259/625)w^3 = 0.4144w^3$. The proof uses no assumption about symmetry or a finite generating set of the unknown body. After normalization, its minimum circumradius divides the argument into two regimes. In the first, a radial divergence estimate is strengthened on eight disjoint contact caps. In the second, four circumsphere contacts determine an explicit inner body. A mixed-volume inequality forces additional volume in every constant-width completion of that inner body. Opposite contact edges are treated jointly: a change of normal coordinates cancels midpoint displacement, a convexity argument controls edge obliquity, and three exact deficit identities couple the resulting integrals. We develop the necessary support-function and mixed-volume preliminaries, derive the boundary formulas, and specify every finite numerical premise and its directed-rounding verification. The code, rational input data, and freshly recomputed reports accompany this manuscript. The result is not the sharp Meissner inequality; the remaining losses are identified at the end.

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1 The theorem and how the proof reaches it

A convex body is a compact convex subset of $\mathbb{R}^3$ with nonempty interior. Its width in a unit direction is the distance between the two supporting planes perpendicular to that direction. A body of constant width w has the same width in every direction. The three-dimensional Blaschke–Lebesgue problem asks for the least volume in this class.

Theorem 1.1. *If $K \subset \mathbb{R}^3$ is a convex body of constant width $w > 0$, then*

$$\text{Vol}(K) > \frac{259}{625}w^3 = 0.4144w^3. \quad (1.1)$$

The proof is computer-assisted. Its numerical premises are the three finite certificates stated and explained in sections 10, 11 and 13.

The conjectured sharp value is the volume of a Meissner tetrahedron,

$$V_M = \frac{\pi}{12} \left(8 - 3\sqrt{3} \arccos \frac{1}{3} \right) = 0.419860045965080 \dots \quad (1.2)$$

at width one; see [3, 4]. We do not assume this conjecture. Hyra’s posted computation gives the lower bound $(130838246407123/10^{15})\pi = 0.411040473721188 \dots$ [2]. We use the radial method of that work, with the full derivation and a new finite set of rational data specified here. The other part of the proof is a geometric completion estimate. Standard foundations are referenced to [1]; all identities needed to connect them to our estimate are derived below.

Dilation multiplies width by w and volume by w^3 . We therefore work at width one until the final step. Translate the center of the smallest containing ball to the origin and call its radius R . We will prove

$$\frac{1}{2} \leq R \leq R_J := \sqrt{3/8}.$$

The exact dividing radius and deficit cutoff are

$$R_* = \frac{60849}{100000}, \quad A_0 = \frac{2}{25}. \quad (1.3)$$

The final coverage is as follows. Decimals in this table are only readable summaries of outward-enclosed rational numbers.

Circumradius	Estimate used	Certified lower value exceeds
$1/2 \leq R \leq 3/5$	Analytic radial estimate	0.4173574
$3/5 \leq R \leq R_*$	Eight rational cap certificates	0.4144109551797
$R_* \leq R \leq R_J$	Contact hull and completion	0.4144467

Here is the reason for the split. Near R_J , the contacts of the smallest sphere are nearly a regular tetrahedron. They provide a useful inner body, but merely containing this inner body is not the full constant-width condition. We must also account for volume that a constant-width completion has to add. Farther from R_J , the radial argument is more effective and does not require the nearly tetrahedral description.

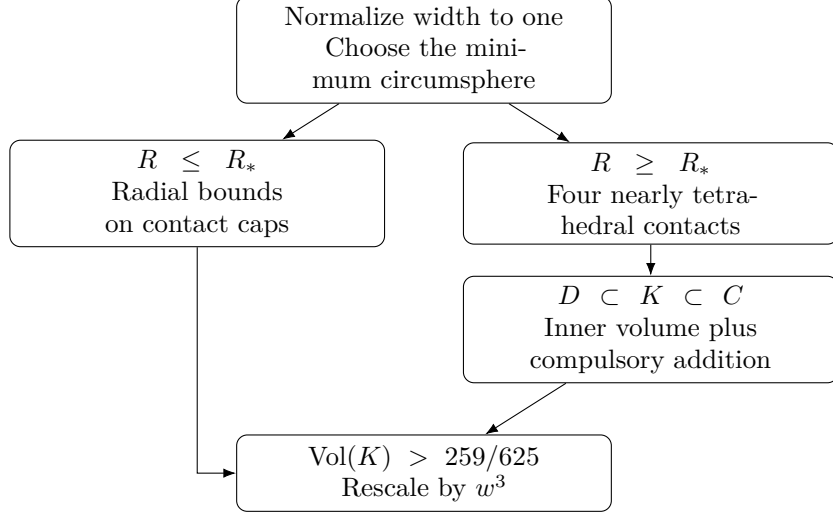


Figure 1: The proof has two complementary geometric branches. The numerical certificates verify explicit inequalities inside these branches; they do not search over unknown convex bodies.

2 Preliminaries: support functions, width, and curvature

2.1 Support functions and containment

For a nonempty compact convex set E its support function is

$$h_E(n) = \max_{x \in E} x \cdot n, \quad n \in \mathbb{S}^2.$$

It extends to all of $\mathbb{R}^3$ by positive homogeneity. The intersection of the half-spaces $x \cdot n \leq h_E(n)$ is exactly E , by separation. Thus

$$E \subset F \iff h_E \leq h_F. \quad (2.1)$$

The Minkowski sum is $E + F = \{x + y : x \in E, y \in F\}$. Maximizing a linear functional separately in x, y proves

$$h_{E+F} = h_E + h_F, \quad h_{tE} = th_E \quad (t \geq 0), \quad h_{-E}(n) = h_E(-n).$$

We write $B = \{x : |x| \leq 1\}$. Its support function is the constant one.

Lemma 2.1 (Width and opposite points). *A convex body has width one if and only if*

$$h_K(n) + h_K(-n) = 1 \quad (n \in \mathbb{S}^2). \quad (2.2)$$

Such a body has diameter one, $K + (-K) = B$, and the following property: if n is an outward unit normal at $x \in \partial K$, then $x - n \in K$.

Proof. The supporting planes have equations $n \cdot x = h_K(n)$ and $n \cdot x = -h_K(-n)$, proving the width assertion and the Minkowski identity. For $x, y \in K$, take $n = (x - y)/|x - y|$ when $x \neq y$. Then $|x - y| \leq h_K(n) + h_K(-n) = 1$. Conversely, opposite supporting planes have points of K at projected distance one, so the diameter is one. Finally, choose $y \in K$ on the plane opposite x . The equations $(x - y) \cdot n = 1$ and $|x - y| \leq 1$ force $x - y = n$. $\square$

The same argument shows that each supporting plane of a constant-width body has a unique contact point. Uniqueness of the contact point does not mean that the boundary has a unique normal at every point.

2.2 The unit-ball transform

For a nonempty compact set E , put

$$\mathcal{B}(E) = \bigcap_{e \in E} \overline{B}(e, 1).$$

This operation reverses inclusion. Also a width-one body obeys

$$K = \mathcal{B}(K). \quad (2.3)$$

Diameter one proves $K \subset \mathcal{B}(K)$. Conversely, if $x \in \mathcal{B}(K)$, choose a point minimizing $n \cdot y$ on K . Then

$$n \cdot x \leq n \cdot y + |x - y| \leq -h_K(-n) + 1 = h_K(n).$$

Use equation (2.1). This simple fact will turn an outer approximation of K into an inner approximation.

2.3 Mixed volumes and the normalization of the coefficients

We use the standard three-dimensional mixed volume $\mathcal{V}(E, F, G)$, symmetric, continuous, monotone in each argument, and multilinear under Minkowski addition and nonnegative scaling. Its normalization is $\mathcal{V}(E, E, E) = \text{Vol}(E)$. These foundational properties and the surface-area representation are described in [1, Chapter 5].

Steiner's formula, in our normalization, reads

$$\text{Vol}(E + tB) = V_E + S_E t + M_E t^2 + \frac{4\pi}{3} t^3, \quad (2.4)$$

where $V_E = \text{Vol}(E)$, S_E is surface area, $S_E = 3\mathcal{V}(E, E, B)$, and $M_E = 3\mathcal{V}(E, B, B)$. In particular M_E is the coefficient of t^2 , not the total integral of mean curvature under a possibly different convention.

Lemma 2.2 (Blaschke's identity). *For a width-one body,*

$$S_K = 2V_K + \frac{2\pi}{3}, \quad M_K = 2\pi. \quad (2.5)$$

Proof. Set $m = \mathcal{V}(K, K, -K)$. Reflection invariance gives $\mathcal{V}(K, -K, -K) = m$. Expanding $B = K + (-K)$ yields

$$\frac{4\pi}{3} = 2V_K + 6m, \quad S_K/3 = V_K + m.$$

Eliminate m . Also $2M_K = 3\mathcal{V}(K + (-K), B, B) = 3\text{Vol}(B) = 4\pi$. □

2.4 Curvature-radius matrices and support-function integrals

The spherical gradient and Hessian are taken with respect to the unit-sphere metric. For a twice differentiable support function h , define

$$\mathcal{Q}h = \nabla_{\mathbb{S}^2}^2 h + hI$$

as a symmetric operator on each two-dimensional tangent plane. The support point is $h(n)n + \nabla h(n)$; differentiating tangentially shows that its radius matrix is $\mathcal{Q}h$. Its eigenvalues are the principal radii of curvature, rather than the principal curvatures.

For a symmetric 2×2 matrix L , the cofactor is $\text{cof } L = (\text{tr } L)I - L$. Define the mixed determinant

$$D_2(L, N) = \frac{1}{2} \text{tr}(\text{cof } L N).$$

Then

$$\det((1-t)L + tN) = (1-t)^2 \det L + 2t(1-t)D_2(L, N) + t^2 \det N. \quad (2.6)$$

The area formula and the mixed-volume formula are

$$S_E = \int_{\mathbb{S}^2} \det \mathcal{Q} h_E d\sigma, \quad 3\mathcal{V}(F, E, E) = \int_{\mathbb{S}^2} h_F \det \mathcal{Q} h_E d\sigma. \quad (2.7)$$

Here $\sigma(\mathbb{S}^2) = 4\pi$.

These formulas extend to the bodies below without assuming smooth boundaries. If $E + F = B$, the distributional matrices $\mathcal{Q}h_E, \mathcal{Q}h_F$ are positive semidefinite measures whose sum is $I d\sigma$. Both are therefore bounded matrix densities between 0 and I . It follows in local coordinates that $h_E, h_F \in C^{1,1}$; that is, their first derivatives are Lipschitz. The area measures in equation (2.7) have the displayed bounded almost-everywhere densities. Convolution by rotations gives smooth support functions, preserves inclusion and $E + F = B$, and converges in C^1 and in $W^{2,p}$ for every finite p . Consequently the identities used here pass to the limit. This also applies to K because $K + (-K) = B$.

Lemma 2.3 (Two variation identities). *Suppose $D \subset K$ are Minkowski summands of B . Put $f = h_K - h_D$, $A = \mathcal{Q}h_D$, and $H = \mathcal{Q}h_K$. Then*

$$3(V_K - V_D) = \int f[\det A + D_2(A, H) + \det H] d\sigma, \quad (2.8)$$

$$2(S_K - S_D) = \int f(\text{tr } A + \text{tr } H) d\sigma. \quad (2.9)$$

Proof. Let $E_t = (1-t)D + tK$. Multilinearity gives $\frac{d}{dt}V_{E_t} = \int f \det((1-t)A + tH) d\sigma$. Integrate equation (2.6) over $t \in [0, 1]$; all three coefficients integrate to $1/3$. For area, differentiation and the mixed-area representation give $\frac{d}{dt}S_{E_t} = \int f \text{tr}((1-t)A + tH) d\sigma$. One can also obtain the last equality by integrating the cofactor expression by parts; $\text{div cof}(\mathcal{Q}h) = 0$ on the sphere. Integration in t proves equation (2.9). The smoothing argument above covers the $C^{1,1}$ case. $\square$

3 The minimum circumsphere and its contact tetrahedron

3.1 Existence and balance

A smallest closed ball containing K exists by compactness. Put its center at zero, write its radius as R , and let $\mathcal{C} = K \cap R\mathbb{S}^2$ be its contact set. Then $0 \in \text{conv } \mathcal{C}$. To prove this, otherwise choose a unit vector e and $\alpha > 0$ with $e \cdot y \geq \alpha$ on all contacts. The compact subset $\{y \in K : e \cdot y \leq \alpha/2\}$ lies strictly inside the sphere, with maximal radius $\mu < R$ when it is nonempty. For sufficiently small $t > 0$, its points have $|y - te| \leq \mu + t < R$, and the remaining points satisfy $|y - te|^2 \leq R^2 - \alpha t + t^2 < R^2$. This contradicts minimality.

Caratheodory's theorem, or affine dependence elimination in $\mathbb{R}^3$, gives a minimal subset $y_i = Ru_i$, $2 \leq n \leq 4$, with

$$\lambda_i > 0, \quad \sum_i \lambda_i = 1, \quad \sum_i \lambda_i y_i = 0. \quad (3.1)$$

A minimal such subset is affinely independent. The diameter inequalities and balance imply

$$y_i \cdot y_j \geq R^2 - \frac{1}{2}, \quad 0 \geq R^2 - \frac{1}{2}(1 - \lambda_i).$$

Thus

$$\lambda_i \leq 1 - 2R^2, \quad \frac{1}{2} \leq R \leq R_J = \sqrt{3/8}. \quad (3.2)$$

The lower bound uses diameter one. The upper bound follows by summing the weight inequalities and using $n \leq 4$. If $R^2 > 1/3$, three contacts cannot have weights summing to one; hence exactly four are needed. Their convex hull is a tetrahedron containing zero in its interior.

The width identity also gives

$$(1 - R)B \subset K \subset RB, \quad (3.3)$$

since $h_K \leq R$ implies $h_K \geq 1 - R$. In particular the circumcenter is an interior point and the radial function of K is well defined.

3.2 The total squared-edge deficit

For four contacts define

$$d = \frac{3}{8} - R^2, \quad e_{ij} = 1 - |y_i - y_j|^2 \geq 0, \quad A = \sum_{i < j} e_{ij}.$$

We keep A for this scalar sum and use bold symbols for curvature matrices.

Lemma 3.1. *The contacts satisfy*

$$\boxed{16d \leq A \leq \frac{16d}{1 - 8d}}. \quad (3.4)$$

For $R \geq R_*$, in particular, $A < A_0 = 2/25$.

Proof. Expansion using equation (3.1) gives $\sum_j \lambda_j |y_i - y_j|^2 = 2R^2$. Hence

$$\lambda_i = \frac{1}{4} + 2d - \sum_{j \neq i} \lambda_j e_{ij}.$$

Sum in i to obtain

$$8d = \sum_{i < j} (\lambda_i + \lambda_j) e_{ij}.$$

The other two weights are each at most $1/4 + 2d$, so $\lambda_i + \lambda_j \geq 1/2 - 4d$. This proves the upper bound. For the lower bound set $\bar{y} = \frac{1}{4} \sum_i y_i$. The identity $\sum_{i < j} |y_i - y_j|^2 = 4 \sum_i |y_i - \bar{y}|^2$ gives

$$R^2 = \frac{3}{8} - \frac{A}{16} + |\bar{y}|^2. \quad (3.5)$$

Thus $A \geq 16d$. The rational comparison $16(3/8 - R_*^2)/(1 - 8(3/8 - R_*^2)) < 2/25$ gives the last assertion. $\square$

3.3 Quantitative geometry without assuming symmetry

We need to know that the sphere intersections keep the intended incidence. Here is a convenient quantitative justification. Choose the centered regular unit tetrahedron with columns

$$t_0 = \frac{(1, 1, 1)}{2\sqrt{2}}, \quad t_1 = \frac{(1, -1, -1)}{2\sqrt{2}}, \quad t_2 = \frac{(-1, 1, -1)}{2\sqrt{2}}, \quad t_3 = \frac{(-1, -1, 1)}{2\sqrt{2}}.$$

For its 3×4 matrix T , $TT^T = I/2$ and $T^TT = \frac{1}{2}(I_4 - \frac{1}{4}\mathbf{1}\mathbf{1}^T)$. Let E have zero diagonal and off-diagonal entries e_{ij} . Centering the squared-distance matrix shows that the centered contacts are congruent to

$$y_i - \bar{y} = (I + Q)^{1/2}t_i, \quad Q = 2TET^T. \quad (3.6)$$

Indeed, the centered Gram matrix is $-\frac{1}{2}J\mathcal{D}J$, where $J = I_4 - \mathbf{1}\mathbf{1}^T/4$ and $\mathcal{D}_{ij} = |y_i - y_j|^2$. Its difference from the regular Gram matrix is $JEJ/2$, which is also T^TQT . This proves equation (3.6) up to an orthogonal map.

The contribution of a single e_{ij} to Q is $2e_{ij}(t_it_j^T + t_jt_i^T)$. Its eigenvalues, on the directions $t_i - t_j, t_i + t_j$ and their perpendicular, are $-e_{ij}, e_{ij}/2, 0$. Therefore

$$-AI \preceq Q \preceq (A/2)I. \quad (3.7)$$

Let $a_i = \sum_{j \neq i} e_{ij}$. Since $|y_i| = R$, the circumcenter equations in the centered regular frame give

$$(I + Q)^{1/2}(-\bar{y}) = -\frac{1}{4} \sum_i a_i t_i.$$

Now $\sum_i a_i t_i = \sum_{i < j} e_{ij}(t_i + t_j)$ and $|t_i + t_j| = 1/\sqrt{2}$. Hence

$$|\bar{y}| \leq \frac{A}{4\sqrt{2}\sqrt{1-A}}. \quad (3.8)$$

These estimates imply nondegeneracy for $A \leq A_0$. All triangular faces are acute, because their squared side lengths belong to $[1 - A_0, 1]$ and $1 < 2(1 - A_0)$. The centered affine tetrahedron contains a ball of radius $\sqrt{1-A}/\sqrt{24}$; equation (3.8) is smaller, since $3A^2 < 4(1-A)^2$. Thus its circumcenter is interior throughout this whole six-dimensional deficit region. Its circumradius is at most

$$\sqrt{3/8}\sqrt{1+A/2} + \frac{A}{4\sqrt{2}\sqrt{1-A}} < 1. \quad (3.9)$$

The final comparisons hold already at $A = A_0$ and are rechecked in `verification/verify_auxiliary.py`. These statements concern all six edge deficits, not a chosen symmetric family.

4 A forced inner body and its complementary outer body

For the selected four contacts let

$$Y = \{y_0, y_1, y_2, y_3\}, \quad C = RB \cap \mathcal{B}(Y), \quad D = \mathcal{B}(C). \quad (4.1)$$

The constraints $Y \subset K$ and $K \subset RB$ give $K \subset C$. Applying the order-reversing transform and equation (2.3) gives

$$\boxed{D \subset K \subset C}. \quad (4.2)$$

Both C and D have interior. The first contains $(1-R)B$; the second contains $(1-R)B$ because $C \subset RB$. The point of D is that it is forced by the four contact locations and R , without knowing the rest of K . It usually does not have constant width.

Lemma 4.1 (Complementarity of intersections of small balls). *Suppose C is a full-dimensional finite intersection of closed balls whose radii r_i satisfy $0 < r_i \leq 1$, and $D = \mathcal{B}(C)$ is nonempty. Then*

$$C + (-D) = B, \quad h_C(n) + h_D(-n) = 1. \quad (4.3)$$

Proof. Fix a support point $x \in C$ with outward unit normal n . An interior point of C is strictly inside every defining ball. The normal cone of this strictly feasible intersection is therefore the cone generated by the active ball normals. Explicitly,

$$n = \sum_i a_i \frac{x - c_i}{r_i}, \quad a_i \geq 0, \quad |x - c_i| = r_i$$

over the active indices. To recall the elementary justification, if n were outside that finitely generated cone, a separating direction, made slightly strict by adding a direction toward an interior point, would decrease all active constraints and increase $n \cdot x$. Small displacement in that direction would contradict support maximality. This proves the cone formula.

Since the active normals are unit, $\sum_i a_i \geq 1$. Expanding the ball inequality at any $z \in C$ gives

$$\frac{x - c_i}{r_i} \cdot (z - x) \leq -\frac{|z - x|^2}{2r_i}.$$

Consequently

$$|z - (x - n)|^2 = 1 + |z - x|^2 + 2n \cdot (z - x) \leq 1 + \left(1 - \sum_i \frac{a_i}{r_i}\right) |z - x|^2 \leq 1.$$

Thus $x - n \in D$. On the other hand, for any $y \in D$, $|x - y| \leq 1$, so $n \cdot x - n \cdot y \leq 1$. The choice $y = x - n$ attains equality. This is $h_C(n) + h_D(-n) = 1$, and equality of support functions proves the Minkowski identity. $\square$

The lemma applies to equation (4.1): the defining radii are one and $R < 1$. It provides substantially more information than $D \subset K \subset C$. In particular both C and D are summands of the unit ball, so all the curvature formulas of section 2 apply.

Lemma 4.2 (Complementary volume and area identities). *With the notation of equation (2.4),*

$$\boxed{V_D = \frac{4\pi}{3} - M_C + S_C - V_C, \quad S_D = 4\pi - 2M_C + S_C.} \quad (4.4)$$

Also $M_C + M_D = 4\pi$.

Proof. The identity $B = C + (-D)$ allows the formal multilinear substitution $-D = B - C$. Expanding the cubic volume polynomial gives

$$V_D = \text{Vol}(B) - 3\mathcal{V}(C, B, B) + 3\mathcal{V}(C, C, B) - V_C.$$

Expanding $3\mathcal{V}(B - C, B - C, B)$ gives the area formula, since simultaneous reflection leaves B invariant. Finally $M_E = \int h_E d\sigma$, so the support identity gives $M_C + M_D = 4\pi$. The formal subtraction is just a convenient notation for rearranging a polynomial identity between actual Minkowski sums; no set difference is being assigned a mixed volume. $\square$

5 The geometry of the circumsphere-clipped intersection

This section justifies exactly which surfaces and edges must be integrated. We work with four balanced contacts and $A \leq A_0$. Initially assume all six e_{ij} are positive; zero deficits will follow by continuity.

5.1 Why the original vertices are removed by clipping

Consider the triangular face opposite y_l . Give its plane an inward unit normal n_l and equation $n_l \cdot y = b_l$. The origin is interior, so $b_l < 0 < n_l \cdot y_l$. Balance gives

$$b_l = -\frac{\lambda_l}{1 - \lambda_l} n_l \cdot y_l.$$

Positivity of the incident deficits in the weight equation makes $\lambda_l < 1 - 2R^2$. As $n_l \cdot y_l \leq R$, this implies

$$b_l > R - \frac{1}{2R}. \quad (5.1)$$

The three vertices of the face lie in RS^2 . Their planar circumcenter is $b_l n_l$ and their circumradius squared is $R^2 - b_l^2$. Hence the two intersections of their unit spheres are

$$x_l^\pm = \left(b_l \pm \sqrt{1 - R^2 + b_l^2} \right) n_l.$$

The plus coefficient is positive. A direct rearrangement shows $|x_l^+| > R$ if and only if equation (5.1) holds. For the minus coefficient, $|x_l^-| > \sqrt{1 - R^2}$ and $x_l^- \cdot y_l < 0$, so $|x_l^- - y_l|^2 > 1$. Thus that intersection is excluded by the fourth ball. There is no triple intersection of unit spheres in RB .

5.2 The two kinds of circular edges

Fix an edge of the contact tetrahedron, of length $s = |y_i - y_j|$. Its midpoint $o = (y_i + y_j)/2$ is perpendicular to $y_i - y_j$, because $|y_i| = |y_j| = R$. The intersection of the two unit spheres is a circle with

$$r = \sqrt{1 - s^2/4}, \quad m = |o| = \sqrt{R^2 - s^2/4}. \quad (5.2)$$

Write a point on it as $x = o + rv$, $v \perp (y_i - y_j)$ and $|v| = 1$. The clipping condition $|x| \leq R$ is

$$v \cdot \frac{-o}{m} \geq \frac{1 - s^2/2}{2rm}.$$

Thus the surviving short arc is centered in direction $-o$ and has angular length

$$\phi = 2\beta, \quad \beta = \arccos \frac{1 - s^2/2}{2rm}. \quad (5.3)$$

It is nondegenerate because

$$4r^2m^2 - (1 - s^2/2)^2 = R^2(4 - s^2) - 1 \geq \frac{1}{8} - \frac{3A_0}{16} > 0,$$

using $R^2 \geq 3/8 - A/16$ from equation (3.5). We call these six arcs the *unit-unit edges*.

There are also edges where a unit sphere meets RS^2 . Put

$$\kappa = \sqrt{1 - \frac{1}{4R^2}}, \quad \gamma = \arccos \frac{1}{2R}, \quad d_R = R - \frac{1}{2R}. \quad (5.4)$$

The intersection of $S(y_i, 1)$ and RS^2 is the circle of radius κ with center $d_R y_i / R$. On that circle another ball, centered at y_j , removes an angular interval of length 2Ψ , where

$$\Psi = \arccos \frac{(1 - 2R^2)s}{4R\kappa m}. \quad (5.5)$$

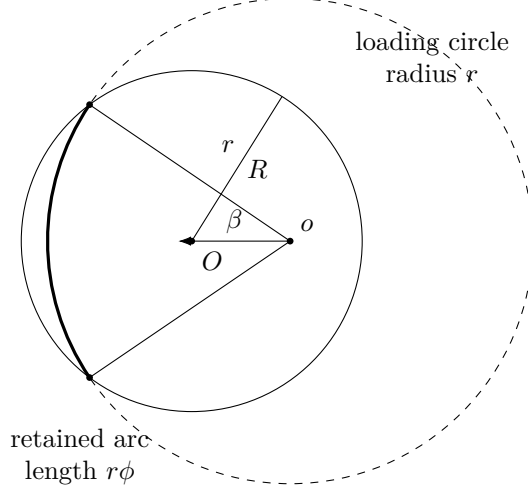


Figure 2: A section in the loading-circle plane, perpendicular to $y_i - y_j$. The solid circle is the circum-ball section and the dashed circle is the intersection of the two unit spheres. The thick inward arc survives clipping. The displayed scale is illustrative; equations (5.2) and (5.3) are exact.

For a derivation, write $x = d_R y_i / R + \kappa v$, $v \perp y_i$, $|v| = 1$. The projection of y_j onto $y_i^\perp$ has length sm/R . The inequality $|x - y_j| \leq 1$ becomes

$$v \cdot \frac{\text{proj}_{y_i^\perp} y_j}{sm/R} \geq \frac{d_R s}{2\kappa m}.$$

The right side is negative. The excluded interval, centered in the opposite direction, has half-angle $\arccos(-d_R s / (2\kappa m))$, which is equation (5.5). We call the surviving arcs the *unit- R edges*.

5.3 Why the same boundary description holds throughout the domain

The un-clipped four-ball intersection has tetrahedral incidence at regularity. Along any path in the positive deficit simplex, equations (3.7) and (3.9) rule out a degenerate contact tetrahedron or tangent triples of unit spheres. The preceding argument excludes a unit-sphere triple meeting RB , while equation (5.3) excludes tangency of two unit spheres with the R -sphere. Thus the incidence of the clipped arcs cannot change.

For a starting configuration take a uniformly shrunken regular tetrahedron, of side $l < 1$ close to one and circumradius $\sqrt{3/8}l$. The inward midpoint of a clipped loading arc has squared distance

$$1 + l^2/2 - \sqrt{2}l\sqrt{1 - l^2/4} < 1$$

from either of the other two centers. No third sphere crosses the arc, so its whole relative interior is in the other balls. Connectedness and the absence of crossing events extend this conclusion to the full domain. Equivalently, the forbidden caps on RS^2 overlap pairwise but have no triple interior intersection. The surviving regions on RS^2 are four spherical triangles.

The boundary consequently has the following pieces:

Piece	Number	Supporting surface
Hexagonal faces	4	Unit spheres
Triangular faces	4	Sphere of radius R
Unit–unit edges	6	Circles of radius r
Unit– R edges	12	Circles of radius κ
Vertices	12	Two unit spheres and the R -sphere

The Euler check is $12 - 18 + 8 = 2$. Zero deficits may collapse some of the new arcs. Shrinking all contact positions slightly makes all deficits positive, retains balance, and converges back to the original configuration. Continuity of support functions, volumes, and the displayed formulas includes all such boundary cases. No generic-position assumption is made in the final theorem.

6 Derivation of the clipped volume formulas

We now compute the two terms V_C, V_D explicitly. The calculation is local along edges, which is why it separates into six scalar functions.

6.1 Angles at a clipped vertex

Let $x \in S(y_i, 1) \cap S(y_j, 1) \cap RS^2$. Its three outward unit normals are

$$n_i = x - y_i, \quad n_j = x - y_j, \quad n_R = x/R.$$

Their scalar products are

$$n_i \cdot n_j = 1 - s^2/2, \quad n_i \cdot n_R = n_j \cdot n_R = 1/(2R).$$

Projecting the two other normals into the tangent plane of a face gives its exterior corner angle. Thus each unit face has corner angle U , and the R -face has angle V , where

$$U = \arccos \frac{s}{4Rr\kappa}, \quad V = \arccos \left(1 - \frac{s^2}{2\kappa^2} \right). \quad (6.1)$$

For example, on the unit face the cosine is

$$\frac{(1/(2R))(1 - (1 - s^2/2))}{\sqrt{1 - (1 - s^2/2)^2} \sqrt{1 - 1/(4R^2)}} = \frac{s}{4Rr\kappa}.$$

This also fixes which supplementary angle is intended.

Let Φ_R be the total angular length of the unit– R arcs. Start with four complete circles, of total angular length 8π . For each pair ij , the forbidden interval $2\Psi_{ij}$ is removed from both circles. There are no overlapping removals on a single circle, by the no-triple-cap property. Hence

$$\Phi_R = 8\pi - 4 \sum_{i < j} \Psi_{ij}. \quad (6.2)$$

6.2 Gauss–Bonnet with all signs and multiplicities

For a disk-like region on a sphere of radius a , with positively oriented piecewise smooth boundary, Gauss–Bonnet reads

$$\frac{\text{area}(F)}{a^2} + \int_{\partial F} k_g \, ds + \sum \text{exterior corner angles} = 2\pi.$$

The geodesic curvature k_g is signed according to the interior of the region. This is especially important on the R -sphere, where one of the signed coefficients below is negative.

Let S_1 be the total area of the four unit faces, and S_R that of the four R -faces. At the two endpoints of edge ij , the two unit faces each contribute U , giving $4U$ in the sum over all faces. The two R -corners give $2V$. A unit–unit edge contributes geodesic-curvature integral $(s/2)\phi$ on each of its two unit faces, hence $s\phi$ in total. A unit– R edge of angular length ω contributes $\omega/(2R)$ on its unit face. On the R -face the signed contribution is $(1 - 1/(2R^2))\omega$: its supporting plane has signed height d_R , and $d_R/R = 1 - 1/(2R^2)$. Consequently

$$S_1 = 8\pi - 4 \sum U - \sum s\phi - \frac{\Phi_R}{2R}, \quad (6.3)$$

$$S_R = R^2 \left[8\pi - 2 \sum V - \left(1 - \frac{1}{2R^2} \right) \Phi_R \right]. \quad (6.4)$$

Every sum over an unlabeled edge quantity in this section is over the six unordered contact pairs $i < j$.

6.3 Volume from boundary vector areas

The divergence theorem gives $3V_C = \int_{\partial C} x \cdot n \, dA$. On a unit face centered at y_i , $x = y_i + n$, so its contribution is its area plus $y_i \cdot \int n \, dA$. On an R -face it is R times its area. The vector-area identity

$$\int_F n \, dA = \frac{1}{2} \int_{\partial F} n \times \, dn \quad (6.5)$$

holds on a unit sphere. It follows, for instance, componentwise from Stokes' theorem applied to the constant-coordinate area forms. This is the same boundary-integral mechanism used for ball-polyhedron volumes in [3]; the following edge contributions are derived for the present clipped geometry.

For a unit–unit edge let $k = (y_j - y_i)/s$, write $x = o + rv$, and choose the orientation $\dot{v} = k \times v$. On the face centered at y_i , $n = (s/2)k + rv$, and

$$\int n \times \, dn = (s/2)k \times \Delta x + r^2 k \phi.$$

The neighboring face has the reverse boundary orientation and normal $-(s/2)k + rv$. Multiplying each vector area by its sphere center and adding leaves

$$-\frac{1}{2}sr^2\phi + \frac{1}{2}\det(y_i, y_j, \Delta x).$$

The endpoints on the R -sphere satisfy

$$\det(y_i, y_j, \Delta x) = 2srm \sin(\phi/2) = s\sqrt{R^2(4 - s^2) - 1} =: T_e.$$

The sign is positive for the orientation just fixed; the short inward arc makes this explicit in the loading-circle plane.

For a unit- R edge, use axis $k = y_i/R$ and write its unit-face normal as $n = -k/(2R) + \kappa v$. The orientation with the unit face on the left is the negative angular orientation about k . Dotting equation (6.5) with $y_i = Rk$ gives $-R\kappa^2\omega/2$. Summing these contributions proves

$$3V_C = S_1 + RS_R - \frac{1}{2} \sum sr^2\phi + \frac{1}{2} \sum T_e - \frac{1}{2} R\kappa^2\Phi_R. \quad (6.6)$$

6.4 The quadratic Steiner coefficient

For a piecewise smooth convex boundary, the t^2 coefficient in its outer parallel volume consists of the integrated mean curvature of its smooth faces and half the exterior angle times edge length. A spherical face of radius a contributes its area divided by a . Along a unit-unit edge, the angle between outward normals is $2\arcsin(s/2)$, so its contribution is $r\phi\arcsin(s/2)$. Along a unit- R edge it is γ , giving $\kappa\omega\gamma/2$. Vertex neighborhoods first contribute to the coefficient of t^3 . Thus

$$M_C = S_1 + \frac{S_R}{R} + \sum r\phi\arcsin(s/2) + \frac{1}{2}\kappa\gamma\Phi_R. \quad (6.7)$$

The coefficients can also be obtained by integrating the normal wedge along an edge in the outer parallel body. This verifies the factor $1/2$, which is important for our numerical normalization.

6.5 The six one-edge functions

Define

$$F(s) = \frac{s}{2} - \frac{s^3}{24} - \sqrt{1 - s^2/4}\arcsin(s/2), \quad (6.8)$$

and

$$z_R = R^2 - R - R^3/3, \quad b_R = \frac{1}{2} - \frac{3}{8R} - \frac{\kappa\gamma}{2} - z_R.$$

For $e = 1 - s^2$ set

$$B_D(R) = \frac{8\pi}{3} - \frac{3\pi}{R} - 4\pi\kappa\gamma, \quad B_C(R) = \frac{8\pi}{3} - \frac{\pi}{R}, \quad (6.9)$$

$$\begin{aligned} f_D(R, e) &= \frac{4}{3}U - 2z_RV + \phi F(s) - \frac{T_e}{6} - 4b_R\Psi, \\ f_C(R, e) &= \frac{1}{3} \left[-4U - 2R^3V + \left(-\frac{3s}{2} + \frac{s^3}{8} \right) \phi + \frac{T_e}{2} + \left(4R^3 + \frac{3}{2R} \right) \Psi \right]. \end{aligned} \quad (6.10)$$

Here all the angles and radicals are exactly those in equations (5.2) to (5.5) and (6.1).

Proposition 6.1 (Separable volume formula). *For every contact tetrahedron in the stated near-regular range,*

$$\boxed{V_D = B_D(R) + \sum_{i < j} f_D(R, e_{ij}), \quad V_C = B_C(R) + \sum_{i < j} f_C(R, e_{ij}).} \quad (6.11)$$

Proof. Insert equations (6.2) to (6.4) into equation (6.6). The coefficient of Φ_R in $3V_C$ simplifies to

$$-\frac{1}{2R} - R^3 \left(1 - \frac{1}{2R^2}\right) - \frac{R\kappa^2}{2} = -R^3 - \frac{3}{8R}.$$

The constant after substituting $\Phi_R = 8\pi - 4\sum \Psi$ is $8\pi - 3\pi/R$. This gives B_C, f_C . Next use $V_D = 4\pi/3 - M_C + S_1 + S_R - V_C$ and equation (6.7). The coefficients of U, V, T_e, Φ_R are respectively $4/3, -2z_R, -1/6, b_R$. The coefficient of ϕ is

$$\frac{s}{3} + \frac{sr^2}{6} - r \arcsin(s/2) = F(s).$$

Substitution of equation (6.2) gives B_D, f_D . □

The separability is geometrically useful: at fixed R , an edge enters only through its own squared length. It does not mean that six arbitrary lengths and an arbitrary R are realizable as circumsphere contacts. Later we extend the formulas analytically to a rectangle solely to obtain safe derivative bounds, and then apply them only to actual contacts.

7 Volume that every constant-width completion must add

The inner volume V_D is not the whole estimate. We now use constant width to force additional volume, without choosing the unknown body K . Set

$$\begin{aligned} A(n) &= \mathcal{Q}h_D(n), & H(n) &= \mathcal{Q}h_K(n), & f(n) &= h_K(n) - h_D(n), \\ g(n) &= 1 - h_D(n) - h_D(-n) = h_C(n) + h_C(-n) - 1. \end{aligned} \tag{7.1}$$

Because $D \subset K$, $f \geq 0$. Constant width gives

$$f(n) + f(-n) = g(n) \geq 0. \tag{7.2}$$

The matrices A, H are contractions almost everywhere, by section 2. The two tangent planes at $n, -n$ are identified by the antipodal isometry whenever their matrices are compared.

The even function g measures the width still missing from D . Equation (7.2) says that a completion must supply this missing support at one or both of the opposite directions. Its distribution is unknown, but an even lower bound for the curvature weight can be used without knowing that distribution.

Proposition 7.1 (Weighted antipodal completion bound). *Define*

$$d_*(n) = \min\{\det A(n), \det A(-n)\}, \quad \psi(s) = \frac{31}{32}s + \min\{1/32, s/2\}.$$

Then

$$\boxed{V_K \geq \frac{47V_D - V_C}{46} + \frac{4}{23} \int_{\mathbb{S}^2} g(n) \psi(d_*(n)) d\sigma(n).} \tag{7.3}$$

Proof. Let $\Delta V = V_K - V_D$ and $c_D = 2V_D + 2\pi/3 - S_D$. By Blaschke's identity, $S_K - S_D = 2\Delta V + c_D$. Subtract $1/32$ times equation (2.9) from equation (2.8). Rearrangement gives

$$\frac{23}{8}\Delta V = \frac{c_D}{16} + \int f \left[\det A - \frac{\text{tr } A}{32} + \det H + \text{tr} \left(\left(\frac{1}{2} \text{cof } A - \frac{1}{32} I \right) H \right) \right] d\sigma. \tag{7.4}$$

We next lower-bound the square bracket for every allowed matrix $\mathbf{H}$. Let $a_1 \leq a_2$ be the eigenvalues of $\mathbf{A}$ and $s = a_1 a_2$. In an eigenbasis of $\mathbf{H}$, its determinant is bilinear in its eigenvalues and the remaining term is linear. Minimizing its orientation first and its two eigenvalues in $[0, 1]$ second reduces to $\mathbf{H} = 0$, a rank-one projection, or $\mathbf{H} = I$. The corresponding additional values are

$$0, \quad a_1/2 - 1/32, \quad 1 + \text{tr } \mathbf{A}/2 - 1/16.$$

The last is positive, so the minimum is $\min\{0, a_1/2 - 1/32\}$. Furthermore

$$a_1 + a_2 \leq 1 + a_1 a_2 = 1 + s, \quad a_1 \geq s.$$

The first inequality is $(1 - a_1)(1 - a_2) \geq 0$. Thus the bracket in equation (7.4) is at least

$$s - \frac{1+s}{32} + \min\{0, s/2 - 1/32\} = \psi(s) - \frac{1}{16}.$$

The function ψ is increasing. Since $f \geq 0$,

$$\int f \psi(\det \mathbf{A}) \geq \int f \psi(d_*) = \frac{1}{2} \int g \psi(d_*), \quad \int f = \frac{1}{2} \int g.$$

Using this in equation (7.4) gives

$$\frac{23}{8} \Delta V \geq \frac{c_D}{16} - \frac{1}{32} \int g + \frac{1}{2} \int g \psi(d_*). \quad (7.5)$$

By equation (7.1), $\int g = 2M_C - 4\pi$. Also equation (4.4) gives

$$2c_D - \int g = 2(V_D - V_C).$$

Substitute and multiply by $8/23$. The result is equation (7.3). All steps were inequalities for every genuine K ; no minimizing curvature field or unproved independence assumption was imposed on K . $\square$

For the integration below we use the polynomial

$$q(s) = \frac{47}{32}s - 2s^2 \leq \psi(s) \quad (s \geq 0). \quad (7.6)$$

Indeed $s/2 - 2s^2 \leq s/2$ and its maximum is $1/32$. On $[0, 1/5]$, q is nonnegative and increasing. The reason for using this particular minorant, rather than evaluating ψ directly, is that it also permits the convexity argument in section 9.2.

8 Opposite contact edges and their common-normal coordinates

There are three partitions of the four contact labels into two pairs: $(01|23)$, $(02|13)$, and $(03|12)$. Fix one, denoted $(ij|kl)$, and set

$$o_1 = (y_i + y_j)/2, \quad o_2 = (y_k + y_l)/2, \quad w = o_2 - o_1,$$

$$k_1 = \frac{y_j - y_i}{|y_j - y_i|}, \quad k_2 = \frac{y_l - y_k}{|y_l - y_k|}, \quad a = e_{ij} + e_{kl}, \quad b = A - a.$$

The two scalars a, b are, respectively, the sum of the opposite-edge deficits and the sum of the four cross-edge deficits. They satisfy $a, b \geq 0$ and $a + b = A \leq A_0$.

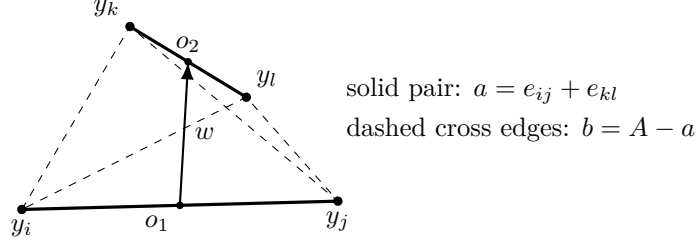


Figure 3: One opposite-edge partition, drawn schematically. The three such partitions satisfy $\sum_j a_j = A$ and $\sum_j b_j = 2A$. They cannot be assigned independent worst-case deficit budgets.

8.1 Exact vector relations and safe scalar bounds

Expansion of squared distances gives

$$|w|^2 = \frac{1}{2} + \frac{a-b}{4}, \quad |k_1 \cdot k_2| \leq c := \frac{b}{2(1-a)}, \quad |w \cdot k_i| \leq \frac{b}{4\sqrt{1-a}}. \quad (8.1)$$

Here are the cancellations behind the estimates. For $u = y_j - y_i$ and $v = y_l - y_k$, the alternating sum of the four cross squared distances is $\pm 2u \cdot v$. The constant terms all cancel, leaving absolute value at most b . Since $|u|, |v| \geq \sqrt{1-a}$, the middle inequality follows. Differences of two cross-distance sums similarly give $4|w \cdot u| \leq b$ and $4|w \cdot v| \leq b$. Finally the sum of all four cross squared distances is $4|w|^2 + |u|^2 + |v|^2$; it equals $4 - b$, whereas $|u|^2 + |v|^2 = 2 - a$. This proves the first identity.

Set

$$z = \frac{k_1 \times k_2}{|k_1 \times k_2|}, \quad w \cdot z > 0, \quad v_1 = z \times k_1, \quad v_2 = z \times k_2.$$

The common normal z exists because $c \leq 1/25 < 1$. It is not generally parallel to w . The contact balance projected onto z implies $o_1 \cdot z < 0 < o_2 \cdot z$; the contacts in each pair have the same projection, and their positively weighted average is zero.

Also $o_i \perp k_i$ and

$$|o_i|^2 = R^2 - |y_i - y_j|^2/4 \geq 1/8 - A/16.$$

Projecting o_1 onto the basis z, v_1 , and using $o_1 \cdot k_2 = -w \cdot k_2$, gives

$$|o_1 \cdot v_1| \leq \frac{b}{4\sqrt{1-a}\sqrt{1-c^2}}.$$

The same holds for $o_2 \cdot v_2$. Thus the angles between z and $-o_1/|o_1|, o_2/|o_2|$ are at most $\arcsin \eta$, where

$$\eta = \frac{b}{4\sqrt{1-a}\sqrt{1-c^2}\sqrt{1/8 - (a+b)/16}}. \quad (8.2)$$

8.2 A common region on which both circle supports are valid

The half-angle of each clipped circle is given by equation (5.3). On our range its cosine increases with s^2 and decreases with R . For example, differentiating its square with respect to s^2 has the sign of $2 - 6R^2 + R^2 s^2 > 0$; this is positive already at the relevant endpoints. Using $s^2 \leq 1$ and $R^2 \geq 3/8 - A/16$ therefore gives

$$\cos \beta \leq C_\beta := \sqrt{\frac{4}{3(2-a-b)}}.$$

Put $S_\beta = \sqrt{1 - C_\beta^2}$ and

$$T(a, b) = \frac{S_\beta \sqrt{1 - \eta^2} - C_\beta \eta}{C_\beta \sqrt{1 - \eta^2} + S_\beta \eta}. \quad (8.3)$$

This is $\tan(\arccos C_\beta - \arcsin \eta)$. For $a, b \geq 0$, $a + b \leq 2/25$, elementary bounds give

$$c \leq 1/25, \quad \eta < 1/16, \quad \frac{1}{2} < T \leq \frac{1}{\sqrt{2}} < \frac{3}{4}. \quad (8.4)$$

For instance $C_\beta \leq 5/6$, and $\tan(\arccos(5/6) - \arcsin(1/16)) > 1/2$ is checked after inserting the radicals $\sqrt{11}, \sqrt{255}$. The upper bound follows from $\arccos C_\beta \leq \arccos \sqrt{2/3}$ and $\eta \geq 0$.

Use coordinates on the hemisphere $n \cdot z > 0$:

$$n = \frac{z + m}{\sqrt{1 + |m|^2}}, \quad m \perp z, \quad x = m \cdot v_1, \quad y = m \cdot v_2. \quad (8.5)$$

Let $\chi = k_1 \cdot k_2 = v_1 \cdot v_2$. The Gram matrix of v_1, v_2 is $\begin{pmatrix} 1 & \chi \\ \chi & 1 \end{pmatrix}$. Inverting it yields

$$W_\chi = 1 + |m|^2 = 1 + \frac{x^2 + y^2 - 2\chi xy}{1 - \chi^2}, \quad d\sigma = \frac{dx dy}{\sqrt{1 - \chi^2} W_\chi^{3/2}}. \quad (8.6)$$

The second formula combines the planar linear Jacobian $1/\sqrt{1 - \chi^2}$ with the central-projection Jacobian $(1 + |m|^2)^{-3/2}$.

Projection of n into $k_1^\perp$ is proportional to $z + xv_1$, and projection into $k_2^\perp$ to $z + yv_2$. Hence on $|x|, |y| < T$ each projection makes angle at most $\arctan T$ with z . Equation (8.3) ensures that it lies strictly inside the clipped loading arc around $-o_1$; the antipodal projection lies inside the other arc around $-o_2$. These points lie in the remaining two unit balls by section 5.

One condition remains: a circle candidate is the support point of the intersection of its two balls only when its normal lies in their normal cone. In the present notation this is

$$|n \cdot k_i| < s_i/2. \quad (8.7)$$

It follows either by resolving n into a nonnegative combination of the two unit normals $\pm(s_i/2)k_i + r_i v_i$, or by that two-dimensional cone's half-angle. We will use the circle formulas only where a positive curvature lower bound proves equation (8.7). Elsewhere the integral estimate will use zero, so no unsupported support formula is assumed.

8.3 Curvature on a circular edge

For an active circle of center o , radius r , and axis k , the support function on its normal region is

$$h(n) = o \cdot n + r \sqrt{1 - (n \cdot k)^2}.$$

Its curvature-radius matrix has eigenvalues

$$0, \quad \frac{r}{\sqrt{1 - (n \cdot k)^2}}. \quad (8.8)$$

To verify this, take the homogeneous function $o \cdot \xi + r \sqrt{|\xi|^2 - (\xi \cdot k)^2}$ and restrict its Euclidean Hessian to $n^\perp$. The linear term has zero Hessian; the norm of the projection into $k^\perp$ has one nonzero tangential eigenvalue as displayed. Complementarity gives $\mathcal{Q}h_D(-n) = I - \mathcal{Q}h_C(n)$, so its determinant is one minus the nonzero radius in equation (8.8).

9 The joint completion integral

9.1 Pairing normals removes the transverse midpoint error

For this opposite pair let r_1, r_2 be its circle radii, and write $w = w_z z + w_\perp$. Whenever both circle supports are valid,

$$g(n) = \frac{r_1 \sqrt{1+x^2} + r_2 \sqrt{1+y^2} - w_z - w_\perp \cdot m}{\sqrt{W_\chi}} - 1. \quad (9.1)$$

The two determinants relevant to d_* are

$$1 - \frac{r_1 \sqrt{W_\chi}}{\sqrt{1+x^2}}, \quad 1 - \frac{r_2 \sqrt{W_\chi}}{\sqrt{1+y^2}}.$$

Define

$$\begin{aligned} r_0 &= \sqrt{3/4}, & r_a &= \sqrt{3/4 + a/4}, & w_0 &= \sqrt{1/2 + (a-b)/4}, \\ q_- &= \min(x^2, y^2), & q_+ &= \max(x^2, y^2), \\ N &= r_0 \sqrt{1+q_+} + r_a \sqrt{1+q_-} - w_0, & \zeta &= \frac{r_a}{\sqrt{1+q_-}}. \end{aligned} \quad (9.2)$$

The common determinant lower bound is $1 - \zeta \sqrt{W_\chi}$. If it is positive, then

$$1 - (n \cdot k_i)^2 > r_i^2 = 1 - s_i^2/4,$$

which proves the required normal-cone condition equation (8.7).

At fixed x^2, y^2 , the two radii obey $r_i = \sqrt{3/4 + e_i/4}$ with $e_1 + e_2 = a$. Their weighted sum in equation (9.1) is concave in e_1 . Its minimum occurs at an endpoint; the smaller endpoint pairs the larger radius with the smaller factor. Therefore

$$r_1 \sqrt{1+x^2} + r_2 \sqrt{1+y^2} \geq r_0 \sqrt{1+q_+} + r_a \sqrt{1+q_-}. \quad (9.3)$$

Also $w_z \leq |w| = w_0$.

The square $[-T, T]^2$, the measure factor, and the determinant lower bound are invariant under $(x, y) \mapsto (-x, -y)$. The term $w_\perp \cdot m$ is odd under this map and cancels exactly. We do not replace it by its absolute value. The average of the two true values of g is nonnegative, since $D \subset K$; hence its lower bound can be clamped at zero.

For fixed N, ζ define

$$\Phi(W) = \frac{[N/\sqrt{W} - 1]_+ q([1 - \zeta \sqrt{W}]_+)}{W^{3/2}}. \quad (9.4)$$

Here $[t]_+ = \max\{t, 0\}$. On all values used below $W \geq 1 + q_+$, so $\zeta \sqrt{W} \geq \sqrt{3}/2 > 4/5$. Its positive determinant argument is therefore less than $1/5$, where q is increasing and nonnegative. Combining equations (7.6) and (9.3) and the antipodal pairing, the contribution of the positive-normal square to $\int g\psi(d_*)$ is at least

$$\int_{[-T, T]^2} \frac{\Phi(W_\chi)}{\sqrt{1-\chi^2}} dx dy. \quad (9.5)$$

At points with a nonpositive determinant lower bound, this expression is zero. Their actual contribution is nonnegative, so excluding them is valid.

9.2 Convexity averages edge obliquity

Lemma 9.1. *The function equation (9.4) is decreasing and convex for the relevant $W \geq 1 + q_+$.*

Proof. On the interval where both unclamped factors are positive set

$$G = N/\sqrt{W} - 1, \quad D_* = 1 - \zeta\sqrt{W}, \quad u = \sqrt{W}/N \in (0, 1), \quad v = \zeta\sqrt{W} \in [4/5, 1).$$

All three factors $G, q(D_*), W^{-3/2}$ are nonnegative and decreasing, which proves monotonicity. For convexity write

$$\phi_1 = GD_*W^{-3/2}, \quad \phi_2 = GD_*^2W^{-3/2}, \quad \Phi = \frac{47}{32}\phi_1 - 2\phi_2.$$

Two differentiations give $\phi_i'' = NW^{-4}B_i(u, v)$, with

$$B_1 = 6 - \frac{15}{4}(u + v) + 2uv,$$

$$B_2 = 6 - \frac{15u}{4} - \frac{15v}{2} + 4uv + 2v^2 - \frac{3}{4}uv^2.$$

The minimum of the bilinear B_1 on $[0, 1]^2$ is $1/2$. The coefficient of u in $B_1/2 - B_2$ is $15/8 - 3v + 3v^2/4 < 0$ for $4/5 \leq v \leq 1$. Its minimum is therefore at $u = 1$, where it equals

$$\frac{-9 + 21v - 10v^2}{8} \geq \frac{7}{40}.$$

Thus $\phi_2'' \leq \phi_1''/2$ and $\Phi'' \geq (15/32)\phi_1'' > 0$. Once one of the positive factors vanishes, Φ stays zero; its derivative jumps upward to zero, preserving convexity. If there was no positive interval, the conclusion is immediate. Finally $W_\chi - 1 - x^2 = (y - \chi x)^2/(1 - \chi^2) \geq 0$, and likewise for y , which verifies the domain used throughout. $\square$

At fixed $|x|, |y|$, the two signs of xy give W_+, W_- . Their average is

$$\bar{W} = 1 + \frac{x^2 + y^2}{1 - \chi^2} \leq W_c := 1 + \frac{x^2 + y^2}{1 - c^2}.$$

Convexity and then monotonicity imply

$$\Phi(W_+) + \Phi(W_-) \geq 2\Phi(\bar{W}) \geq 2\Phi(W_c). \quad (9.6)$$

As $1/\sqrt{1 - \chi^2} \geq 1$, equation (9.5) is bounded below by

$$\mathcal{F}(a, b) = 4 \int_0^{T(a, b)} \int_0^{T(a, b)} \frac{[N/\sqrt{W_c} - 1]_+ q([1 - \zeta\sqrt{W_c}]_+)}{W_c^{3/2}} dx dy. \quad (9.7)$$

The remaining geometric obliquity enters only through c^2 . In contrast, replacing χxy by its absolute worst value in each quadrant would introduce a first-order loss.

9.3 Summing the three pairs without overlap

Proposition 9.2. *For the three opposite-edge partitions and their deficits a_j, b_j ,*

$$V_K \geq \frac{47V_D - V_C}{46} + \frac{8}{23} \sum_{j=1}^3 \mathcal{F}(a_j, b_j), \quad (9.8)$$

where

$$\sum_{j=1}^3 a_j = A, \quad \sum_{j=1}^3 b_j = 2A. \quad (9.9)$$

Proof. Each positive determinant region corresponds to an interior support point of a specified unit–unit edge of C . The clipped-arc and normal-cone conditions are strict there. Distinct edges cannot share such a normal: a finite intersection of strictly convex balls is strictly convex, so it has a unique support point, and an interior point of the present edge has exactly its two unit balls active. Thus these regions are disjoint except on null boundaries. Each pair and its antipode contribute at least $2\mathcal{F}(a_j, b_j)$ to the integral of equation (7.3), proving equation (9.8). Each of the six deficits belongs to exactly one opposite pair, proving equation (9.9). $\square$

10 First finite certificate: the two-parameter completion integral

The first numerical premise is small in dimension: the unknown body has been eliminated, leaving the explicit function equation (9.7) of two nonnegative numbers. We state the exact rational target, then explain its verification.

Lemma 10.1 (Certified affine lower bound). *For $a, b \geq 0$ and $a + b \leq 2/25$,*

$$\boxed{\mathcal{F}(a, b) > L(a, b) := \frac{153}{100000} - \frac{601}{100000}a + \frac{317}{20000}b.} \quad (10.1)$$

10.1 A fixed integration square

The integrand of equation (9.7) is symmetric in x, y . On $0 \leq y \leq x \leq T$ put $x = Ts$, $y = Tst$, with $0 \leq s, t \leq 1$. Its planar Jacobian is T^2s , so

$$\mathcal{F}(a, b) = 8T^2 \int_0^1 \int_0^1 s \Phi(W_c(Ts, Tst)) ds dt. \quad (10.2)$$

There is no minimum/maximum switch between x^2 and y^2 in these coordinates: $x^2 \geq y^2$. Define the two smooth factors

$$G = \frac{r_0\sqrt{1+x^2} + r_a\sqrt{1+y^2} - w_0}{\sqrt{W_c}} - 1, \quad D_* = 1 - \frac{r_a\sqrt{W_c}}{\sqrt{1+y^2}}.$$

The complete integrand equals $8T^2s [G]_+ q([D_*]_+) W_c^{-3/2}$. It is nonnegative and locally Lipschitz in the parameters. On a region where both factors are positive it is an elementary smooth function of a, b, s, t involving only rational operations and square roots.

10.2 Validated quadrature at a parameter center

For fixed rational a, b , partition $[0, 1]^2$ into 192^2 equal cells. A cell is retained only if directed interval arithmetic proves $G > 0$ and $D_* > 0$ throughout it. All other cells are omitted. This yields a valid lower integral because the true integrand is nonnegative.

Let P denote the smooth complete integrand on a retained cell, m its center, and $h = 1/384$ its half-side. The midpoint estimate is

$$\frac{1}{|Q|} \int_Q P \geq P(m) - \frac{h^2}{6} \left(\sup_Q |P_{ss}| + \sup_Q |P_{tt}| \right). \quad (10.3)$$

For clarity, in one variable Taylor's formula with integral remainder gives $|\frac{1}{2h} \int_{-h}^h p(t) dt - p(0)| \leq h^2 \sup |p''|/6$. Apply this successively to the two variables and bound each pure second derivative on the full rectangle. No mixed derivative is required. The right side of equation (10.3) is clamped below by zero, then multiplied by the cell area and summed with downward rounding. This produces $\underline{\mathcal{F}(a, b)}$.

10.3 Control between parameter centers

On a parameter rectangle, a separate 64^2 interval partition in s, t encloses $\mathcal{F}_a, \mathcal{F}_b$. The derivatives of $T(a, b)$ and the Jacobian factor $8T^2s$ are included. On a cell where either positive factor is known to be nonpositive, its derivative contribution is zero. On an active cell one uses the ordinary smooth derivative. A possibly switching cell uses the interval hull of zero and the smooth derivative enclosure. This is valid almost everywhere for the locally Lipschitz positive-part function, and hence under integration. Equivalently, apply the fundamental theorem of calculus along a parameter segment and then Fubini's theorem. This argument does not differentiate twice across a positive-part boundary.

For a parameter rectangle of center m and half-widths h_a, h_b , the acceptance test is

$$\underline{\mathcal{F}(m)} - L(m) - h_a \sup |\mathcal{F}_a - L_a| - h_b \sup |\mathcal{F}_b - L_b| > 0. \quad (10.4)$$

When it holds, the whole rectangle satisfies equation (10.1).

10.4 Coverage and completed output

The initial partition is the 32×32 dyadic tiling of $[0, 2/25]^2$. A rectangle is excluded only if its lower endpoints have sum strictly larger than $2/25$. Otherwise it is tested by equation (10.4); an inconclusive rectangle is bisected in the longer direction, with a deterministic tie rule. The recomputation gives

Accepted parameter rectangles	584
Rectangles excluded by the exact triangle test	474
Attempted parameter rectangles	618
Maximum dyadic depth in either parameter	6
Point-quadrature cells per center	192^2
Derivative-quadrature cells per rectangle	64^2

The independent Python cover checker reconstructs the same tree. It verifies that every accepted leaf is used exactly once and every untested leaf is outside the triangle. Thus summing the areas is not used as a substitute for a genuine cover check.

The least acceptance margin at 48-bit dyadic precision is

$$\frac{17153402}{2^{48}} > 0. \quad (10.5)$$

The 52-bit recomputation gives $274686139/2^{52} > 0$ and the same 584 leaves. Both runs were completed for this manuscript. Arithmetic details and the source-to-equation correspondence appear in section 15. These finite checks, together with the preceding error and coverage proofs, prove theorem 10.1.

Corollary 10.2. *The three opposite-edge contributions satisfy*

$$\sum_{j=1}^3 \mathcal{F}(a_j, b_j) > \frac{459}{100000} + \frac{2569}{100000} A. \quad (10.6)$$

Consequently, for $R \geq R_*$,

$$V_K > \frac{47V_D - V_C}{46} + \frac{8}{23} \left[\frac{459}{100000} + \frac{2569}{100000} 16(3/8 - R^2) \right]. \quad (10.7)$$

Proof. Sum equation (10.1) and use equation (9.9): $-601/100000 + 2(317/20000) = 2569/100000 > 0$. Then use $A \geq 16(3/8 - R^2)$ and equation (9.8). $\square$

This calculation explains why the opposite-edge budgets must remain coupled. They cannot all take an independent minimum of $\mathcal{F}$ at unrelated values of a, b .

11 Second finite certificate: the inner–outer volume combination

Define from equations (6.9) and (6.10)

$$\mathfrak{f}(R, e) = \frac{47f_D(R, e) - f_C(R, e)}{46}, \quad H_0(R) = \frac{47B_D(R) - B_C(R)}{46} + 6\mathfrak{f}(R, 0). \quad (11.1)$$

Then

$$\frac{47V_D - V_C}{46} = H_0(R) + \sum_{i < j} [\mathfrak{f}(R, e_{ij}) - \mathfrak{f}(R, 0)]. \quad (11.2)$$

When $R < R_J$, the zero-deficit term $H_0(R)$ is an analytic baseline, not the volume of a realizable regular unit tetrahedron at radius R . Only the complete expression equation (11.2) is applied to geometry.

Lemma 11.1 (Verified derivative signs). *Throughout $R_* \leq R \leq R_J$, $0 \leq e \leq A_0$,*

$$H'_0(R) > 0, \quad \mathfrak{f}_e(R, e) > 0, \quad \mathfrak{f}_{ee}(R, e) < 0, \quad \mathfrak{f}_{eR}(R, e) < 0. \quad (11.3)$$

11.1 Derivatives in a cancellation-resistant form

The certificate checks the signs of exact derivatives, not finite differences. Here are the expressions that make those checks practical. Put $x = 1 - e$, $Q = R^2(4 - x) - 1$, $r = \sqrt{1 - x}/4$, and $\theta = \arcsin(\sqrt{x}/2)$. Keeping the angle notation from section 6, differentiation gives

$$\begin{aligned} U_x &= -\frac{1}{\sqrt{x}(4-x)\sqrt{Q}}, & V_x &= \frac{R}{\sqrt{x}\sqrt{Q}}, \\ \Psi_x &= -\frac{R(1-2R^2)}{\sqrt{x}(4R^2-x)\sqrt{Q}}, & \phi_x &= \frac{2(6R^2-R^2x-2)}{(4-x)(4R^2-x)\sqrt{Q}}, \\ (T_e)_x &= \frac{4R^2-1-2R^2x}{2\sqrt{x}\sqrt{Q}}, & F_x &= -\frac{\sqrt{x}}{16} + \frac{\theta}{8r}. \end{aligned} \quad (11.4)$$

For example, differentiate each cosine, use $(\arccos z)' = -z'/\sqrt{1-z^2}$, and simplify by $4r^2m^2 - (1 - x/2)^2 = Q$. Each denominator is positive on the rectangle. Substitution into equation (6.10) cancels the larger angle derivatives and gives

$$\partial_x f_C(R, 1-x) = \frac{(2-x)\sqrt{Q}}{4\sqrt{x}(4R^2-x)} - \frac{(4-x)\phi}{16\sqrt{x}}, \quad (11.5)$$

$$\begin{aligned} \partial_x f_D(R, 1-x) &= \phi \left(-\frac{\sqrt{x}}{16} + \frac{\theta}{8r} \right) + \frac{(x+6-8R)\sqrt{Q}}{4\sqrt{x}(4R^2-x)} \\ &\quad - \frac{(6R^2-R^2x-2)\theta}{\sqrt{4-x}\sqrt{Q}(4R^2-x)} - \frac{\sqrt{4R^2-1}(1-2R^2)\gamma}{\sqrt{x}\sqrt{Q}(4R^2-x)}. \end{aligned} \quad (11.6)$$

The code differentiates the combination of these expressions in x and R . Recall $\partial_e = -\partial_x$: hence the three required edge signs are equivalent to $\mathfrak{f}_x < 0$, $\mathfrak{f}_{xx} < 0$, $\mathfrak{f}_{xR} > 0$ in the x notation.

11.2 The complete derivative cover

Divide $[R_*^2, 3/8]$ into 32 rational intervals and $[0, A_0]$ into 100 rational intervals. Taking an outward square root in the first coordinate gives 3,200 boxes in (R, e) . On each box the program encloses equations (11.5) and (11.6) and the exact first derivatives of their combination. It also encloses H'_0 on all 32 radius intervals. All four strict comparisons in equation (11.3) pass at both 128-bit and 192-bit fixed-dyadic precision. Every radical and inverse-trigonometric argument range is checked before evaluation. This proves theorem 11.1 subject only to the explicit arithmetic implementation described below.

11.3 Turning the signs into a six-edge estimate

The positive, concave edge increments can be bounded by secants; no optimization over the six edge lengths remains. Partition $[R_*^2, 3/8]$ into 512 consecutive rational intervals $[z_j, z_{j+1}]$. Put

$$d_j = 3/8 - z_j, \quad E_j = \frac{16d_j}{1 - 8d_j}, \quad m_j = \frac{f(\sqrt{z_{j+1}}, E_j) - f(\sqrt{z_{j+1}}, 0)}{E_j} > 0.$$

Each $E_j > 0$, including the last interval's left endpoint. For $\sqrt{z_j} \leq R \leq \sqrt{z_{j+1}}$, the contact budget gives $0 \leq e_{ij} \leq E_j$. Since $f_{eR} < 0$, integrating in e first gives

$$f(R, e) - f(R, 0) \geq f(\sqrt{z_{j+1}}, e) - f(\sqrt{z_{j+1}}, 0).$$

Concavity in e bounds the latter below by its chord $m_j e$. Hence

$$\frac{47V_D - V_C}{46} \geq H_0(\sqrt{z_j}) + m_j A \geq H_0(\sqrt{z_j}) + 16d_{j+1}m_j.$$

Combining with equation (10.7), every body in that interval satisfies

$$V_K > H_0(\sqrt{z_j}) + 16d_{j+1}m_j + \frac{8}{23} \left[\frac{459}{100000} + \frac{2569}{100000} 16d_{j+1} \right]. \quad (11.7)$$

The 512 right sides are evaluated with directed intervals and compared with 259/625 by integer arithmetic. At 128-bit precision the smallest lower endpoint is

$$\frac{141028929484742528892496879275454481331}{2^{128}} > 0.4144467 > \frac{259}{625}. \quad (11.8)$$

The 192-bit run verifies the same strict target. Thus the complete near-Jung range is controlled. Neither a sampled tetrahedron nor an assumed worst symmetric configuration enters this conclusion.

12 A radial inequality for the complementary circumradii

We now give the other branch of the proof. The method is due to Hyra [2]; it is derived here to specify exactly what the rational certificate must verify. In this section assume temporarily that ∂K is C^1 . The last section removes this assumption.

12.1 Radial function and the opposite-point inequality

Since $(1 - R)B \subset K$, define the continuous radial function

$$\rho(u) = \max\{t \geq 0 : tu \in K\}, \quad 1 - R \leq \rho(u) \leq R.$$

Polar coordinates give

$$3V_K = \int_{\mathbb{S}^2} \rho(u)^3 d\sigma(u). \quad (12.1)$$

At a boundary point x with outward unit normal n , theorem 2.1 gives $x - n \in K \subset RB$. Squaring gives

$$2x \cdot n \geq |x|^2 + 1 - R^2. \quad (12.2)$$

Take $0 < m \leq 1 - R^2$ and the radial vector field $F_m(x) = 2x/(|x|^2 + m)$. Its boundary flux obeys $F_m(x) \cdot n \geq 1$, so the divergence theorem yields

$$S_K \leq \int_K \operatorname{div} F_m dx.$$

At radius r , $\operatorname{div} F_m = 2(r^2 + 3m)/(r^2 + m)^2$. The radial integral is explicit because

$$\frac{d}{dr} \frac{2r^3}{r^2 + m} = \frac{2r^2(r^2 + 3m)}{(r^2 + m)^2}.$$

Therefore

$$S_K \leq \int_{\mathbb{S}^2} \frac{2\rho^3}{\rho^2 + m} d\sigma. \quad (12.3)$$

12.2 A tangent majorant and its recoverable deficit

Let $s_0 > 0$ and define

$$\begin{aligned} a &= \frac{2(s_0^2 + 3m)}{3(s_0^2 + m)^2}, & b &= \frac{2s_0^3}{s_0^2 + m} - as_0^3, \\ G(s) &= as^3 + b - \frac{2s^3}{s^2 + m}. \end{aligned} \quad (12.4)$$

These constants are the slope and intercept of the tangent line at $X = s_0^3$ to $\phi(X) = 2X/(X^{2/3} + m)$. Direct differentiation gives

$$\phi''(X) = -\frac{4(X^{2/3} + 5m)}{9X^{1/3}(X^{2/3} + m)^3} < 0.$$

Thus $G(s) \geq 0$, with equality only at $s = s_0$. Also G decreases below s_0 and increases above s_0 , since its derivative has the sign of $\phi'(s_0^3) - \phi'(s^3)$. Using equations (12.1) and (12.3) and Blaschke's identity gives

$$2V_K + 2\pi/3 \leq 3aV_K + 4\pi b - \int G(\rho) d\sigma.$$

Whenever $3a - 2 > 0$,

$$V_K \geq \frac{2\pi/3 - 4\pi b + \int_{\mathbb{S}^2} G(\rho) d\sigma}{3a - 2}. \quad (12.5)$$

The aim of the cap construction is to retain a positive portion of the last integral, rather than discard it everywhere.

For $R \leq 3/5$, choose $m = 16/25$ and $s_0 = 37/80$ and discard it. The conditions $m \leq 1 - R^2$ and $3a - 2 > 0$ hold. Exact rational simplification yields

$$V_K \geq \frac{76439431}{575385750}\pi > 0.4173574 > \frac{259}{625}. \quad (12.6)$$

It remains to cover $3/5 \leq R \leq R_*$.

12.3 Angular separation of the four contacts

For a radius interval $R_- \leq R \leq R_+$ in this range, put

$$\eta = \frac{1}{2R_-^2} - 1.$$

The contact directions $u_i = y_i/R$ obey $u_i \cdot u_j \geq -\eta$. They also obey the upper bound

$$u_i \cdot u_j \leq T_* := \frac{4\eta^2}{1-\eta} - 1. \quad (12.7)$$

For a proof, let $t = u_1 \cdot u_2$ and $A' = \lambda_1 + \lambda_2$. Dot the balance equation with u_1, u_2 separately, replace cross scalar products by their lower bound, and add:

$$A'(1+t) \leq 2\eta(1-A').$$

For the other two labels the same estimate, using their mutual scalar product at least $-\eta$, gives $(1-A')(1-\eta) \leq 2\eta A'$. Thus $A' \geq (1-\eta)/(1+\eta)$. Substitution proves equation (12.7). Taking the larger η from R_- weakens the bound and makes it valid on the whole radius interval.

12.4 Upper radial bounds near the negative contact directions

If u is at angle $\theta < \pi/2$ from $-u_i$, the points $\rho(u)u$ and Ru_i have distance at most one. Hence

$$\rho(u)^2 + 2R \cos \theta \rho(u) + R^2 \leq 1.$$

Replacing R by R_- decreases the left side. Its positive root gives

$$\rho(u) \leq \Pi(R_-, \cos \theta) := -R_- \cos \theta + \sqrt{1 - R_-^2 \sin^2 \theta}. \quad (12.8)$$

This upper bound increases with θ . Therefore a bound at the outer edge of a cap also bounds all its inner directions.

12.5 Lower radial bounds propagated from the positive contacts

Set $m = 1 - R_+^2$ for the radius interval. The normal-angle inequality equation (12.2) says that, at a point of radial value s , its outward normal makes angle at most

$$\gamma_m(s) = \arccos \frac{s^2 + m}{2s}$$

with the radial direction. This is real on the actual radial range and increases with s , since $s^2 \leq R^2 < m$.

Suppose two directions u, u' are separated by angle δ and the normal at su' , $s = \rho(u')$, makes angle γ with u' . Its supporting plane gives

$$\rho(u) \leq \frac{s \cos \gamma}{\cos(\gamma + \delta)}$$

when $\gamma + \delta < \pi/2$. The ratio on the right increases with γ ; its derivative is $\sin \delta / \cos^2(\gamma + \delta)$. Consequently

$$\rho(u) \leq N_m(\rho(u'), \delta), \quad N_m(s, \delta) = \frac{s(s^2 + m)}{(s^2 + m) \cos \delta - \sqrt{4s^2 - (s^2 + m)^2} \sin \delta}. \quad (12.9)$$

This function increases with s and with δ wherever the denominator is positive. For the first assertion, rewrite it as $[(s^2 + m)/2] / \cos(\gamma_m(s) + \delta)$; both factors increase with s .

Thus, if $N_m(X, \delta) \leq Y$ and the radial value at one direction is at least Y , then the radial value at a direction within δ is at least X . Otherwise monotonicity applied to equation (12.9) would give a strict contradiction. Beginning at $\rho(u_i) = R \geq R_-$ propagates lower bounds outward from every positive contact.

12.6 A rational grid makes every propagation test algebraic

Choose $N = 4096$ and use

$$\theta_k = 2 \arctan(k/N), \quad c_k = \frac{N^2 - k^2}{N^2 + k^2}, \quad s_k = \frac{2Nk}{N^2 + k^2}. \quad (12.10)$$

These are exact rational cosines and sines. For $\delta_k = \theta_k - \theta_{k-1}$, angle subtraction gives rational c_k^Δ, s_k^Δ as well.

Let $X, Y > 0$ be rational candidates and put $T = X^2 + m$, $D_* = 4X^2 - T^2$. The three rational inequalities

$$D_* \geq 0, \quad Yc_k^\Delta - X > 0, \quad Y^2 D_* (s_k^\Delta)^2 \leq T^2 (Yc_k^\Delta - X)^2 \quad (12.11)$$

prove $N_m(X, \delta_k) \leq Y$ and positivity of its denominator. Indeed, both sides being nonnegative, take the square root of the last inequality to get $Y\sqrt{D_*}s_k^\Delta \leq T(Yc_k^\Delta - X)$. Rearranging gives $Tc_k^\Delta - \sqrt{D_*}s_k^\Delta \geq TX/Y > 0$, as required.

A decreasing sequence $w_0 = R_-, w_1, \dots, w_{n_+} > s_0$ passing equation (12.11) with $X = w_k, Y = w_{k-1}$ proves

$$\rho(u) \geq w_k \quad \text{for } d(u, u_i) \leq \theta_k. \quad (12.12)$$

Induction along a shortest great-circle segment proves this also for points strictly inside each cap, since a previous point can be chosen at distance at most δ_k and inside the preceding cap.

For negative caps, a rational $z_k \leq s_0$ is an upper bound at angle θ_k whenever

$$z_k + R_- c_k \geq 0, \quad (z_k + R_- c_k)^2 \geq 1 - R_-^2 s_k^2. \quad (12.13)$$

The first sign makes taking a square root legitimate, so equation (12.13) implies $z_k \geq \Pi(R_-, c_k)$. Equation (12.8) then applies throughout the cap.

12.7 Disjoint caps and the integrated deficit

Require $n_+, n_- < N$ and

$$c_{n_+}^2 - s_{n_+}^2 > T_*, \quad c_{n_-}^2 - s_{n_-}^2 > T_*. \quad (12.14)$$

The half-angles are below $\pi/2$, and equations (12.7) and (12.14) ensure that distinct positive caps are disjoint, as are distinct negative caps. A positive cap and a negative cap cannot meet: their radial inequalities would give both $\rho > s_0$ and $\rho \leq s_0$ there.

The spherical ring $\theta_{k-1} \leq d(u, u_i) \leq \theta_k$ has area $2\pi(c_{k-1} - c_k)$. On a positive ring, $G(\rho) \geq G(w_k)$, and on a negative ring $G(\rho) \geq G(z_k)$, by the monotonicities following equation (12.4). The remaining directions contribute nonnegatively. Thus, defining

$$\Delta = 4 \sum_{k=1}^{n_+} (c_{k-1} - c_k) G(w_k) + 4 \sum_{k=1}^{n_-} (c_{k-1} - c_k) G(z_k),$$

we have $\int G(\rho) \geq 2\pi\Delta$. The resulting bound is

$$V_K \geq \pi \frac{2/3 - 4b + 2\Delta}{3a - 2}. \quad (12.15)$$

This is the sole numerical radial expression checked in the next section.

13 Third finite certificate: the radial intervals

The eight consecutive intervals below cover $[3/5, R_*]$ exactly. Every number in the first three columns is a terminating decimal interpreted as an exact rational. In every row $N = 4096$, and all candidates w_k, z_k are integer multiples of 10^{-12} stored in `verification/data/radial_candidates.json`.

R_-	R_+	s_0	n_+	n_-	Lower endpoint, approximately
.60000	.60500	.479322	796	1675	.416450260822
.60500	.60700	.485500	758	1794	.415235053685
.60700	.60800	.489000	735	1850	.414578330679
.60800	.60825	.492000	714	1889	.414557800923
.60825	.60840	.492500	711	1897	.414452404230
.60840	.60845	.492900	708	1902	.414436432226
.60845	.608475	.493000	707	1903	.414421400194
.608475	.60849	.493000	707	1904	.414410955179

The checker verifies the endpoint partition, $m = 1 - R_+^2$, $3a - 2 > 0$, the strict ordering and ranges of the candidates, every inequality in equations (12.11) and (12.13), and both disjointness comparisons in equation (12.14). These tests are rational inequalities. No trigonometric function is numerically evaluated on the angular grid.

Each nonnegative ring contribution to Δ is rounded downward to an integer multiple of 10^{-18} before summation. This weakens the bound in the correct direction. The rational coefficient of π in equation (12.15) is positive. We use the independently proved lower bound

$$\pi_{40} = 6 \sum_{k=0}^{39} \frac{\binom{2k}{k}}{(2k+1)2^{4k+1}} < \pi. \quad (13.1)$$

It is the first 40 terms of the positive series $6 \arcsin(1/2) = \pi$. Multiplying by π_{40} gives a rational volume lower bound. Every one of the eight results exceeds $259/625$ by integer cross-multiplication. The same rational lower bound for π verifies equation (12.6).

The full rational result is stored in the machine-readable report. For a short exact bound suitable for the text, that result is strictly greater than

$$\frac{4144109}{10000000} = 0.4144109 > \frac{259}{625}. \quad (13.2)$$

The full endpoint is approximately 0.41441095517972193. The additional comparison with the compact rational in equation (13.2) is checked by integer cross-multiplication in the manuscript audit script. The rational candidates may originally be proposed by floating-point search, but the default verification does not run that search and does not trust it. An inaccurate candidate fails an exact test rather than being accepted on the basis of its origin.

14 Assembly for all constant-width bodies

Proof of theorem 1.1. For a width-one body with C^1 boundary, choose its minimum circumsphere. If $R \leq 3/5$, use equation (12.6). If $3/5 \leq R \leq R_*$, use the eight radial certificates. If $R_* \leq R \leq R_J$, use equations (11.7) and (11.8). These ranges cover every possible circumradius. Every branch has a fixed positive margin above $259/625$, and there are finitely many final radius intervals. In particular the smallest certified value is the full radial endpoint bounded below in equation (13.2).

For an arbitrary width-one convex body set

$$K_\epsilon = \frac{K + \epsilon B}{1 + 2\epsilon}, \quad \epsilon > 0.$$

It has width one. The boundary of $K + \epsilon B$ is C^1 : its outward normal is the normalized displacement from its unique nearest point in the closed convex set K , and nearest-point projection is continuous. Thus the first part applies to every K_ϵ . These bodies converge in Hausdorff distance, and hence in volume, to K . The fixed positive margin above $259/625$ survives the limit. Finally, dilation by w multiplies volume by w^3 , giving equation (1.1). $\square$

Notice that the word “strict” in equation (1.1) is not inferred merely from a sequence of strict inequalities. It follows from the uniform quantitative margin of the finite assembly. The numerical estimates are not asserted to be attained by any body.

15 Arithmetic, reproducibility, and the role of the computer

15.1 What is a premise and what is not

The finite premises of the theorem are exactly:

Finite statement	Mathematical justification for its use
The affine bound equation (10.1)	Validated quadrature, parameter derivative bounds, and the complete cover in section 10.
The derivative signs and 512 values in section 11	The exact edge formulas and the secant argument, applied to the contact budget.
The eight rational radial rows	The propagation, sign, disjointness, and ring-integration implications in section 12.

The programs do not prove the normal-cone lemma, the clipped boundary incidence, the geometric support-region statements, or the mixed-volume identities. Those are the analytic parts proved in this manuscript. Conversely, no unproved numerical optimization statement is used in those analytic arguments. This separation makes the computer-assisted claim auditable.

15.2 Fixed-dyadic intervals

At precision b , an interval is represented as $2^{-b}[L, U]$ with integer endpoints. To enclose a rational p/q , use the floor and ceiling of $2^b p/q$. Addition and subtraction of interval endpoints are exact integer operations. For multiplication compute all four integer endpoint products, take their minimum and maximum, divide by 2^b , and round outward. An interval containing zero cannot be inverted. On an interval of one strict sign, the reciprocal is decreasing, so its endpoints are obtained from $2^{2b}/U$ and $2^{2b}/L$ with outward rounding.

A square crossing zero has lower endpoint zero and upper endpoint the larger squared endpoint, rounded upward. For square roots, the lower integer endpoint is the integer square root of $L2^b$; the upper is the integer square root of $U2^b$, increased by one unless its square is exact. These rules are inclusion-preserving by elementary order properties. Any negative radicand or zero denominator stops verification.

The Python hull arithmetic uses arbitrary-size integers at $b = 128$ and $b = 192$. The C++ integral arithmetic uses 64-bit endpoints at $b = 48$ and $b = 52$, with all products in signed 128-bit integers. A product of signed 64-bit integers fits in signed 128 bits; every conversion back to 64 bits is range-checked. The square-root routine uses a floating root only as an initial integer proposal, then corrects it until exact integer inequalities prove the lower and upper endpoints. No floating-point sign comparison is used to accept a cell.

15.3 Derivative enclosures

A derivative jet stores a function value and its first and second derivatives as intervals. Addition, multiplication, and composition propagate the exact rules, for example

$$(fg)' = f'g + fg', \quad (fg)'' = f''g + 2f'g' + fg'',$$

$$(\phi(f))' = \phi'(f)f', \quad (\phi(f))'' = \phi''(f)(f')^2 + \phi'(f)f''.$$

The integral checker carries two pure second derivatives because equation (10.3) does not require a mixed derivative. The hull checker differentiates the analytic derivative expressions equations (11.5) and (11.6). It does not differentiate a table of sampled values or a finite difference approximation.

15.4 Enclosing the inverse trigonometric functions

The hull formulas require inverse trigonometric values. For $0 \leq x < 7/10$ we use

$$\arcsin x = \sum_{n=0}^{\infty} a_n x^{2n+1}, \quad a_n = \frac{\binom{2n}{n}}{4^n(2n+1)}.$$

The coefficients decrease. After retaining $n = 0, \dots, 48$, the positive tail is at most $a_{48}x^{99}/(1-x^2)$. Monotonicity encloses interval arguments by enclosing their endpoints. For $0 \leq x < 1/2$, the alternating arctangent series through $x^{81}/81$ has its remainder between $-x^{83}/83$ and zero. Machin's identity

$$\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$$

therefore gives an independent interval for π .

When an arccosine argument is near one, the implementation uses

$$\arccos x = 2 \arctan \sqrt{\frac{1-x}{1+x}}.$$

Otherwise it uses $\pi/2 - \arcsin x$, with signs handled explicitly. All argument ranges are checked. Derivatives use the exact formulas $(\arcsin x)' = (1-x^2)^{-1/2}$ and $(\arcsin x)'' = x(1-x^2)^{-3/2}$, and their arctangent counterparts, not derivatives of an uncontrolled truncated series.

The small edge function F in equation (6.8) is evaluated without subtraction of nearly equal numbers. With $x = s/2$ it has the positive series

$$F(s) = \sum_{n=1}^{\infty} \frac{\beta_n}{2n+3} x^{2n+3}, \quad \beta_0 = 1, \quad \beta_n = \frac{2n}{2n+1} \beta_{n-1}. \quad (15.1)$$

For a derivation differentiate $x - x^3/3 - \sqrt{1-x^2} \arcsin x$; the derivative is $\sum_{n \geq 1} \beta_n x^{2n+2}$. Integrate from zero. The checker retains $n = 1, \dots, 34$. The first omitted power is s^{73} . For the value and its first two s -derivatives, the ratio between subsequent positive terms is bounded above by

$$\frac{s_+^2}{4} \frac{75 \cdot 74}{73 \cdot 72} < 1.$$

A geometric-series bound encloses all three tails. The jet chain rule then includes the derivatives of s with respect to the chosen variable.

15.5 The files and commands

The directory `verification/` contains all source files and finite input data. Its principal correspondence with this proof is:

File	Purpose
<code>projective_certificate.cpp</code>	equations (10.2) to (10.4) and the adaptive parameter certificate.
<code>verify_auxiliary.py</code>	Geometric constants, the independent cover check, and assembly of radial rows.
<code>clipped_hull.py, combined_hull.py</code>	equations (6.10), (11.1), (11.5) and (11.6).
<code>verify_weighted_hull.py</code>	3,200 derivative boxes and 512 secant lower bounds.
<code>cap_certificate.py,</code> <code>data/radial_candidates.json</code>	All rational propagation and cap data.
<code>interval.py, analytic_functions.py</code>	Directed interval and derivative operations and rigorous series tails.
<code>verify_regular_endpoint.py</code>	Regular tetrahedral specialization and constants.
<code>verify.py</code>	Full execution and the exact final comparison.

From that directory run

```
python verify.py --output reports/recheck_48_128
python verify.py --cpp-bits 52 --interval-bits 192 \
    --output reports/recheck_52_192
```

The requirements are Python 3 and a C++17 compiler supporting signed 128-bit integers, such as `g++`. No third-party Python package is needed for these two commands. They recompile the checker, recompute all cell bounds, reconstruct the cover independently, verify all radial data, and record exact lower endpoints and SHA-256 hashes. Stored reports are not used as numerical premises. Verification refuses Python’s optimization flag `-O`, which would disable assertions.

Both complete commands were executed in preparing this manuscript. Their reports are supplied in `reports/run_48_128/` and `reports/run_52_192/`. They agree on the cover and pass all required inequalities. These are two precisions of the same implementation, not two independently developed mathematical implementations. The result has not been formalized in a proof assistant or independently peer-reviewed.

16 The remaining sharp gap

For exact regular tetrahedral contacts, $R = R_J$ and all deficits vanish. The clipping disappears. Put $\alpha = \arccos(1/3)$. The formulas above give

$$V_C = V_R = \frac{8\pi}{3} + \frac{\sqrt{2}}{4} - \frac{27\alpha}{4}, \quad V_D = 2V_M - V_R.$$

They can be obtained directly by setting the collapsed unit- R edge sum to zero in the surface and volume formulas. At regularity, $S_C = 8\pi - 18\alpha$ and $M_C - S_C = (\pi\sqrt{3}/2)\alpha$, which also verify the dual-volume identity. Using $A = 0$ in equation (10.7) yields the rigorous specialization

$$V_K \geq \frac{47(2V_M - V_R) - V_R}{46} + \frac{459}{287500} > 0.4190589 \quad (16.1)$$

for every width-one body containing the regular unit tetrahedron. Such a body necessarily has $R = R_J$: the tetrahedron already has that circumradius, and equation (3.2) supplies the reverse inequality.

This value remains below V_M . Universally, theorem 1.1 gives only

$$0.4144 < \inf_{\text{width}(K)=1} V_K \leq 0.419860045965080 \dots$$

The relative difference is about 1.30044% of V_M .

The steps that lose sharpness are explicit. The weighted completion estimate minimizes a curvature expression without retaining the differential relation between that curvature and the support increment. The polynomial minorant and restricted circle-support regions weaken it further. The affine bound for $\mathcal{F}$ introduces another numerical loss, although it correctly retains the three coupled budgets. The radial branch also uses a tangent majorant and recovers its deficit only on selected caps. None of these estimates is claimed to be equality-sensitive at a Meissner body.

A proof of the sharp value must repair these losses or replace the corresponding estimates. Improving a finite decimal comparison, by itself, is not such a proof. In particular, no inclusion monotonicity conjecture for Meissner generators and no conjectural minimization over tetrahedron-containing bodies is used in the theorem above.

A Notation at a glance

Symbol	Meaning
K, w, R	Unknown constant-width body, its width, and normalized minimum circumradius.
$B, \mathcal{B}(E)$	Unit ball and intersection of unit balls centered at E .
$Y = \{y_i\}$	Four positively balanced circumsphere contacts.
e_{ij}, A, d	Squared-edge deficits, their sum, and $3/8 - R^2$.
C, D	Clipped outer body $RB \cap \mathcal{B}(Y)$ and its unit-ball transform.
V_E, S_E, M_E	Volume and the two nonconstant Steiner coefficients.
$h_E, \mathcal{Q}h_E$	Support function and curvature-radius matrix.
f, g	Support increment $h_K - h_D$ and missing width $1 - h_D(n) - h_D(-n)$.
A, H	Curvature matrices of D and K , not the scalar deficit A .
s, r, m	One contact edge length, its loading-circle radius, and midpoint distance.
ϕ, U, V, Ψ	The four edge angles defined in section 6.
a, b	Opposite-edge and cross-edge deficit sums for one partition.
χ, c, η, T	Actual edge cosine, its bound, midpoint tilt bound, and projective cutoff.
x, y, W_χ	Common-normal projective coordinates and their normalization factor.
$\mathcal{F}, L$	Explicit completion integral and its certified affine lower bound.
$\mathfrak{f}, H_0$	Weighted one-edge volume term and zero-deficit analytic baseline.
ρ, m, s_0, a, b, G	Radial function and tangent-majorant data; the last four scalar symbols are local to sections 12 and 13.

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